joined by the large subunit of the ribosome and this complex is known as the translation initiation complex.

In bacteria the small ribosomal subunit attaches directly to a pattern of sequence in the mRNA known as Shine-Dalgarno sequences which are present just before start codons and "point them out" to the ribosome.

Since the bacterial mRNA are polycistronic, they are transcribed in groups (called operons), so one bacterial mRNA can contain the coding sequences for several genes. A Shine-Dalgarno sequence marks the start of each coding sequence, letting the ribosome find the right start codon for each gene.

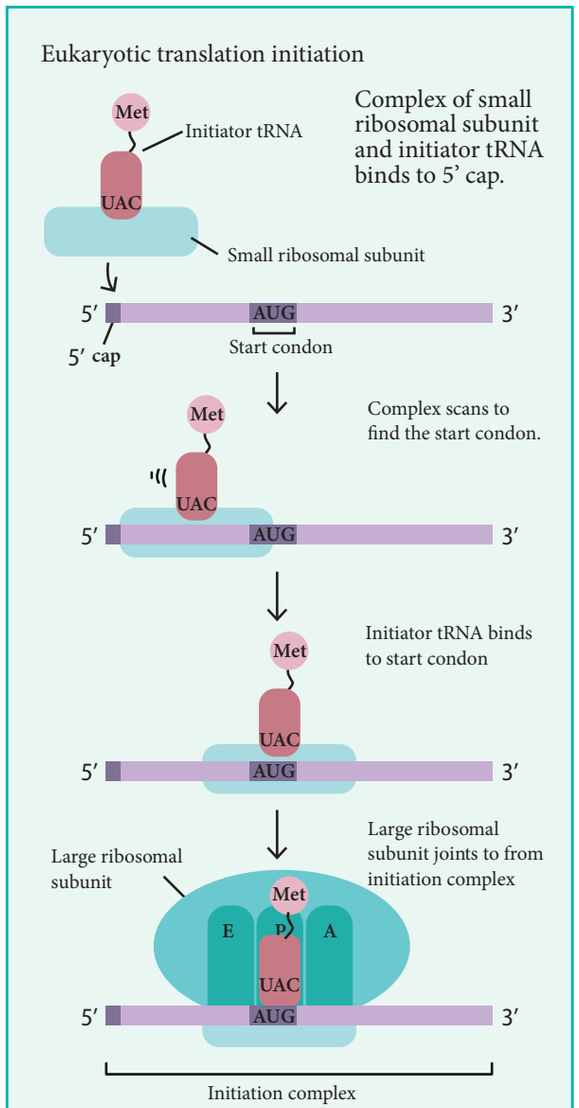


Figure 15 The translation initation process in eukaryotes

6.4.4.2 Translation Elongation

1. The first methionine-carrying tRNA starts in the middle slot or the P site of the ribosome. The subsequent codon is open to the A site.
2. The next tRNA with the anticodon which is complementary to this codon occupies the A site of the ribosome.
3. Once the matching tRNA has landed in the A site the first amino acid methionine from the first tRNA is covalently bonded (peptide bond) to the amino acid of the second tRNA in the A site.
4. Thus a dipeptide is formed with the methionine as N-terminus and the other amino acid is the C-terminus.

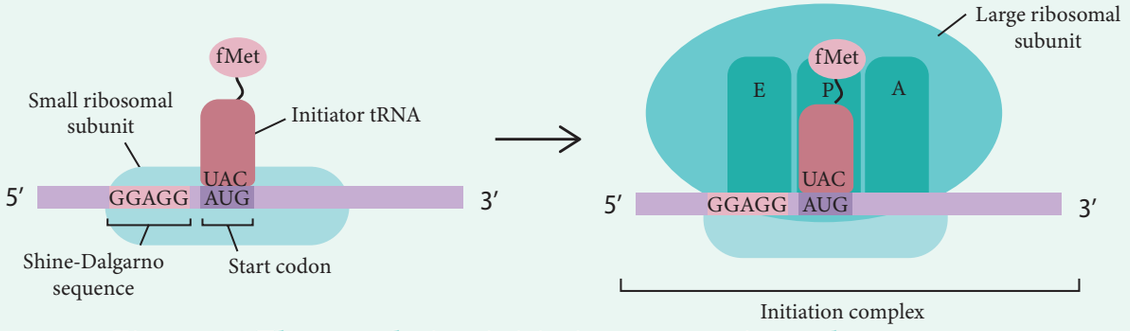


Figure 16 The translation initiation process in prokaryotes

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